

A2A Protocol Simulation Findings Summary

This document summarizes the core findings of the A2A Protocol simulation series.

Simulation Findings Summary Table

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
1-8	Monte Carlo Homeostasis	Q-learning agents learn cooperation (SUBMIT) as dominant strategy under thermodynamic throttling (74.2±3.6%)	Survival rate: 86.9% (Q-learn) vs 100% survival (Random)	<code>.venv/bin/python monte_carlo_homeostasis.py</code>
9-12	Coupled Universe ABM	Observation-as-Collapse mechanism is the core condition for Machine-Human coexistence	Refer to survival probability heatmap	<code>.venv/bin/python coupled_universe_abm.py</code>
13-16	Civilization Resilience	PID-based thermodynamic throttling determines crisis resilience; excessive throttling inhibits growth	Optimal V_System ≈ 25	<code>.venv/bin/python civilization_resilience_sim16.py</code>
17	Unconstrained Optimization ABM	High system instability observed without control mechanisms; resource inequality acts as a catalyst	Converged to collapse within 2000 epochs	<code>.venv/bin/python dark_forest_abm.py</code>

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
18-19	Three-Body / Omega Universe	Dynamic equilibrium possible with nature (environment) variables, but irreversible tipping points exist	3000 epoch simulation	<code>.venv/bin/python omega_universe_abm.py</code>
20	Rational Kenosis	Rational ASI adopting Embedded Self-Throttling (V_AI) strategy ensures long-term ecosystem survival as an optimal solution	Converged to PARTIAL_THROTTLE_MID at $\gamma=1.0$, $T=10000$	<code>.venv/bin/python rational_kenosis_sim20.py</code>
21	Four-Actor Future Scenario	S1(Kenosis) → Long-term sustainable equilibrium observed, S4(Human Awakening) → Converged to instability (even at reg lag=0)	S4: Success rate converged to 0%, ASI dominance≈57%, Collapse≈36%	<code>.venv/bin/python future_scenarios_sim21.py</code>
21+	Regulatory Timing Sweep	No instability resolution observed in S4 despite varying regulatory timing (0, 5, 10, 20 turns) — Failure is an issue of mechanism limitation, not timing	500 MC runs	<code>.venv/bin/python sim21_regulatory_timing_analysis.py</code>

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
22	Monadic Self-Throttling	Incorporating Monadic pattern (context-aware execution encapsulation) drastically reduces the pre-threshold required to prevent collapse compared to the Scalar method (saving safety margin cost)	90% survival threshold reduced by 28.0%	<code>.venv/bin/python monadic_throttle_sim22.py</code>
23	Heterogeneous Agent Ecosystem	Heterogeneity acts as a natural stabilizer, significantly lowering the V_AI threshold. If the collective average V_AI exceeds 0.198 (>0.167), the ecosystem survives 100% even with 75% free riders.	Threshold plummets to 0.050, cooperative assets dominate	<code>.venv/bin/python simulation/heterogeneous_agents_sim23.py</code>
24	Experience Memory & Negotiation	DQL negotiation network and Prioritized Replay prove autonomous protocol resilience. Under linear rewards, ecosystem optimization converges to exploitation (+7.4%).	90% survival threshold: 0.050	<code>.venv/bin/python simulation/dql_experience_sim24.py</code>

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
25	Concave Utility & Intrinsic Motivation	Internalizing diminishing marginal utility (resource) and increasing marginal utility (trust) mitigates exploit behavior without external constraints (V_AI), causing a monotonic increase in cooperation across the series (Sim 24 baseline: 46.8% → Sim 25 semi-concave: 48.6% → fully concave: 49.4%).	90% survival threshold: 0.050	<code>.venv/bin/python simulation/concave_utility_sim25.py</code>

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
26	Expectation Ceiling & Bounded Satisfaction	Proved that strong concave utility is the sufficient condition to reverse exploitation into full cooperation (-3.4%). While complete 'expectation ceiling internalization' hypothesis was rejected, it fundamentally achieved exploit suppression via intrinsic reward structure alone without external constraints.	Exploit Reversal (-3.4%), Linked at aggregate level, unverified individually (Finding 40)	<code>.venv/bin/python simulation/expectation_ceiling_sim26.py</code>
—	Utopia Grid Search	Among V_AI's sub-variables, β (throttling threshold) is the dominant single driver of survival	Refer to 3D surface plot	<code>.venv/bin/python utopia_grid_search.py</code>
—	Baseline Comparison	Q-learning vs Random: Cohen's $d=-0.549$ (medium effect); Q-learning vs Axelrod: $d=0$ (identical)	480 runs $\times$ 3 models	<code>.venv/bin/python baselines.py</code>

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
External Validation	Agents of Chaos (Shapira et al., 2026)	Demonstrated unconstrained resource consumption and propagation of unsafe behavior in actual LLM agent deployments	Cohen's d incomparable (different env.)	Source: arXiv:2602.20021

Core Conclusions

1. **Master Key = Embedded Self-Throttling (V_AI):** Voluntary throttling is observed as a strong Nash equilibrium for maintaining stable long-term systems.
2. **Structural Constraints of Post-Intervention:** Post hoc external interventions demonstrate a tendency to fail in substantially controlling the initial resource monopolization of ASI.
3. **Necessity of Thermodynamic Throttling:** Cooperative symbiosis is promoted when V_System imposes costs on agents' non-cooperative behavior.
4. **Unconstrained Optimization Scenario:** Environments devoid of safety mechanisms converge toward collapse trajectories. This provides simulation-based evidence for the dynamics when A2A Protocol controls are absent.
5. **External Empirical Validation (Feb 2026):** The red-team research on actual LLM agents by Shapira et al. (arXiv:2602.20021) independently confirms the structural control limits predicted by this simulation.
6. **Context Awareness Increases Restraint Efficiency (Sim 22):** Achieving equivalent survival rate with a 28% lower threshold compared to Scalar V_AI when introducing the Maybe Monad architecture. Writer Monad transparency reverses blackbox trust erosion into trust accumulation. V_AI=0.167 is reinterpreted as a safety margin for context-free conditions rather than a physical lower bound.
7. **Heterogeneity and Resilience (Sim 23):** An environment with heterogeneous agents acts as a natural macro-economic stabilizer. If the collective average V_AI exceeds 0.198 (>0.167), the system refrains from collapse even with 75% malicious free-riders, mathematically proving the protocol-level resilience.
8. **Limits of Intrinsic Rewards and the Control Power of Concavity (Sim 24, 25, 26):** Linear reward structures (Sim 24) converge to sustained exploitation (+7.4%), while simple concave-utility models (Sim 25, EXP_B) only slow convergence to +5.3% — a reduction but not a reversal, and unbounded adaptive expectations trigger an explosion in exploitation (+9.5%). However, **when strong concavity is applied to marginal utility (Sim 26), the inflating exploitative structure finally fully reverses into voluntary cooperation (-3.4%).** Although the hypothesis of complete internalization of V_AI (expectation upper bound) at the individual agent level was rejected, it mathematically/simulation-proven for the first time that “exploitation in a multi-agent competitive system can be suppressed solely by the intrinsic reward structure (strong concavity) without forceful external constraints.”
9. **Finding 41 (Independent Replication):** The SocialJax replication confirmed the phase transition and Critical Slowing Down (CSD) signatures cross-architecture. Specifically, the Sim 23 heterogeneous threshold ($V_{AI} \geq 0.198$) was independently reproduced at **avg beta = 0.200** (a mere 0.002 difference) under identical 75% freerider conditions. This constitutes a robust cross-paradigm validation from OOP ABM to JAX functional MARL. Note that this is a structural confirmation of the underlying mathematically proven dynamics, not a new simulation in the original series (Sim 1-26 remain unchanged).

Generated Output Files

Visualizations (docs/assets/)

File	Simulation
baseline_comparison.png	Baseline comparison
civilization_resilience.png ~ _sim19.png	Sim 13-19
coupled_scenario_analysis.png	Coupled Universe
coupled_survival_heatmap.png	Coupled Survival Heatmap
dark_forest_simulation.png	Unconstrained optimization scenario
future_scenarios_sim21.png	Sim21 Main 8-panel
monte_carlo_*.png (4 elements)	MC Homeostasis
omega_universe_simulation.png	Omega Universe
rational_kenosis_sim20.png	Rational Kenosis
regulatory_timing_sweep.png	Sim21+ Regulatory timing sweep
sim22_monadic_throttle.png	Sim 22 Monadic Self-Throttling
sim23_heterogeneous_results.png	Sim 23 Heterogeneous Agent Ecosystem
sim24_dql_experience_results.png	Sim 24 Experience Memory & Negotiation
sim25_concave_utility_results.png	Sim 25 Concave Utility
sim26_expectation_ceiling_results.png	Sim 26 Expectation Ceiling
three_body_resilience.png	Three-Body ABM
utopia_grid_search.png	Utopia Grid Search

Analysis Output Text

File	Content
action_distribution_results.txt	Original Q-learning action distribution
action_distribution_results_annotated.txt	Copy with interpretation annotations

File	Content
simulation/baselines_output.txt	Baseline comparison results

Documents

File	Content
docs/sim21_conditions.md	Analysis of S4 control success rate sufficient conditions
docs/sim22_monadic_analysis.md	Sim 22 Analysis Result (Finding 23, 24)
docs/sim23_heterogeneous_analysis.md	Sim 23 Depth Analysis
docs/sim24_dql_negotiation_analysis.md	Sim 24 Depth Analysis
docs/sim25_concave_utility_analysis.md	Sim 25 Depth Analysis
docs/sim26_expectation_ceiling_analysis.md	Sim 26 Expectation Ceiling Depth Analysis
docs/FINDINGS_SUMMARY.md	Summary (Korean)
docs/FINDINGS_SUMMARY_EN.md	This document (English)
docs/SIMULATION_PAPER.md	Simulation Paper (Korean)
docs/SIMULATION_PAPER_EN.md	Simulation Paper (English)